CONTINUOUS RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

1 PROBABILITY DENSITY FUNCTIONS

If a random variable X is continuous, that is, if its range is an interval on the real line, its probability distribution is called a ***continuous probability distribution***.

Example: The Normal Distribution and Maxwell's Work

When the Scottish physicist **James Clerk Maxwell** analyzed the velocities of gas molecules in equilibrium, he assumed:

- Molecules move equally likely in the **x**, **y**, and **z** directions.
- The velocity components are **independent**.

Under these assumptions, he derived a continuous probability distribution for molecular speeds — the **normal (Gaussian) distribution**, which later became fundamental in both physics and statistics.

Some other examples of continuous probability distributions include:

- **Natural and Environmental Sciences:** amount of rainfall in a day and the concentration of carbon dioxide in the atmosphere.
- **Biology and Health Sciences:** blood pressure and body temperature
- **Social and Behavioral Sciences:** annual household income and time spent studying in a day.
- **Physical and Material Sciences:** sound intensity in decibels and voltage in an electrical circuit.

A continuous probability distribution differs from a discrete probability distribution in several ways. Clearly, since we have uncountably infinite set of possible values, a continuous probability distribution cannot be expressed in tabular form. Instead, one must give an equation or a formula to describe a continuous probability distribution.

The equation used to describe a continuous probability distribution is called a ***probability density function*** (pdf). We will denote a probability density function as $f(x)$. Below, is the formal definition:

Given a continuous random variable X, a ***probability density function*** (pdf) is a function $f(x)$ such that

1. $f(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f(x)dx = 1$

Thus, a probability density function is a function that is nonnegative everywhere, and the total area under the curve is equal to 1. The value of function is assumed to be zero at all values outside of its range.

Note that since in the continuous case sums are replaced by integrals, these two conditions are analogous to the conditions we gave in the discrete case.

It should be intuitively clear that in the continuous case, probabilities should be defined as various integrals of the probability density function (i.e., areas under the pdf). Moreover, it also follows from the definition that for any continuous random variable X and any real number a,

$$P(X = a) = 0$$

Thus, the probability density function is used to compute the probability of the random variable falling ***within a particular range of values***, as opposed to taking on any one value.

In fact, in the continuous case, we have three possible types of probability problems:

I.

$$P(X \leq a) = P(X < a) = \int_{-\infty}^{a} f(x)dx$$

Thus, this probability corresponds to the area under the pdf to the left of a.

II.

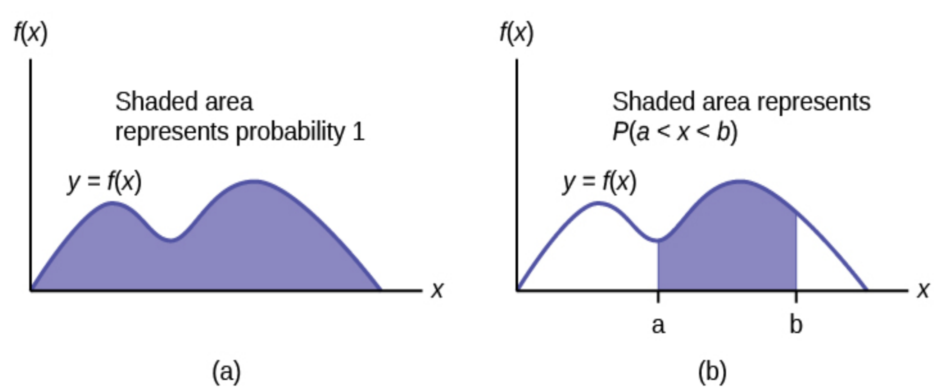
$$P(X \geq a) = P(X > a) = \int_{a}^{\infty} f(x)dx$$

Thus, this probability corresponds to the area under the pdf to the right of a.

III.

$$P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b) = \int_{a}^{b} f(x)dx$$

Thus, this probability corresponds to the area under the pdf between a and b.



Mathematical Note 1:

Recall the basic integration formulas:

(i) **Power rule.** For any real number $r \neq -1$,

$$\int x^r\, dx = \frac{x^{r+1}}{r+1} + C$$

(ii) **Logarithmic rule.**

$$\int \frac{dx}{x} = \ln(x) + C$$

(iii) **Exponential rule.** For any real number $k \neq 0$,

$$\int e^{kx} dx = \frac{1}{k} e^{kx} + C$$

(iv) **Integration by parts.** Let u, v be two functions. The product rule for derivatives tells us

$$d(uv) = udv + vdu$$

Integrating both sides and solving for $\int udv$, we get the integration by parts formula:

$$\int udv = uv - \int vdu$$

Mathematical Note 2:

Recall that

$$\lim_{x\to\infty} e^{-x} = 0$$

In fact,

$$\lim_{x\to\infty} x^n\, e^{-x} = 0$$

for any positive integer n.

EXAMPLES

Example 1. Let X be a continuous random variable and let $f(x) = Ax^4$ in $[0,1]$.

(a) Determine the value of the constant A that makes $f(x)$ a probability density function.

We will make use of the fact that

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Thus, we must have

$$\int_{0}^{1} Ax^4 dx = 1$$

So,

$$A\left[\frac{1}{5}x^5\right]_0^1 = 1$$

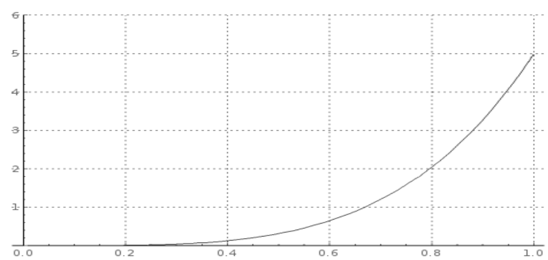
implying

$$\frac{1}{5}A = 1$$

or

$$A = 5$$

The graph of this pdf is given below:



(b) Find $P(X < 0.7)$

$$P(X < 0.7) = \int_{0}^{0.7} 5x^4 dx = [x^5]_0^{0.7} = 0.168$$

(c) Find $P(X > 0.2)$

$$P(X > 0.2) = \int_{0.2}^{1} 5x^4 dx = [x^5]_{0.2}^{1} = 0.999$$

(d) Find $P(0.3 \leq X < 0.6)$

$$P(0.3 \leq X < 0.6) = \int_{0.3}^{0.6} 5x^4 dx = [x^5]_{0.3}^{0.6} = 0.075$$

Example 2. Let X be a continuous random variable and let $f(x) = \frac{A}{x^3}$ in $[1, \infty)$.

(a) Determine the value of the constant A that makes $f(x)$ a probability density function.

We will make use of the fact that

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Thus, we must have

$$\int_{1}^{\infty} \frac{A}{x^3} dx = 1$$

So,

$$\left[-\frac{A}{2x^2}\right]_1^{\infty} = 1$$

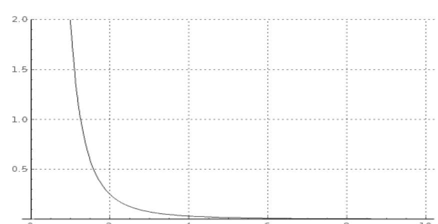
implying

$$\frac{A}{2} = 1$$

or

$$A = 2$$

The graph of this pdf is given below:



(b) Find $P(X < 2)$

$$P(X < 2) = \int_{1}^{2} \frac{2}{x^3} dx = \left[-\frac{1}{x^2}\right]_1^{2} = \frac{3}{4}$$

(c) Find $P(X > 1.5)$

$$P(X > 1.5) = \int_{1.5}^{\infty} \frac{2}{x^3} dx = \left[-\frac{1}{x^2}\right]_{1.5}^{\infty} = \frac{4}{9}$$

(d) Find $P(2 \le X \le 3)$

$$P(2 \le X \le 3) = \int_2^3 \frac{2}{x^3}\,dx = \left[-\frac{1}{x^2}\right]_2^3 = \frac{5}{36}$$

Example 3. Let X be a continuous random variable and let $f(x) = Ae^{-2x}$ in $[0, \infty)$.

(a) Determine the value of the constant A that makes $f(x)$ a probability density function. We will make use of the fact that

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Thus, we must have

$$\int_0^{\infty} Ae^{-2x}dx = 1$$

So,

$$\left[-\frac{A}{2}e^{-2x}\right]_0^{\infty} = 1$$

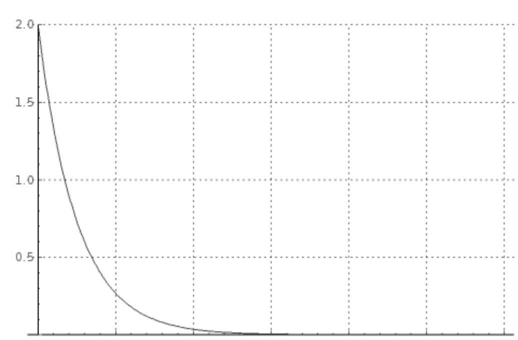
implying

$$\frac{A}{2} = 1$$

or

$$A = 2$$

The graph of this pdf is given below:



(b) Find $P(X < 1)$

$$P(X < 1) = \int_0^1 2e^{-2x}dx = [-e^{-2x}]_0^1 = 1 - e^{-2} = 0.865$$

(c) Find $P(X > 1.5)$

$$P(X > 1.5) = \int_{1.5}^{\infty} 2e^{-2x} dx = [-e^{-2x}]_{1.5}^{\infty} = e^{-3} = 0.05$$

(d) Find $P(1 < X \leq 2)$

$$P(1 < X \leq 2) = \int_{1}^{2} 2e^{-2x} dx = [-e^{-2x}]_{1}^{2} = e^{-2} - e^{-4} = 0.117$$

2 THE CUMULATIVE DISTRIBUTION FUNCTION

Let X be a continuous random variable with probability density function $f(x)$. Then, its ***cumulative distribution function*** (cdf), $F(x)$, is defined in exactly the same way as in the discrete case:

$$F(x) = P(X \leq x)$$

Consequently,

$$F(x) = \int_{-\infty}^{x} f(t)\, dt$$

for any real number x.

Thus, conversely, if F is differentiable at a point x,

$$f(x) = \frac{dF(x)}{dx}$$

Properties

The properties of cumulative distribution functions in the continuous case are exactly the same as those that we listed in the discrete case.

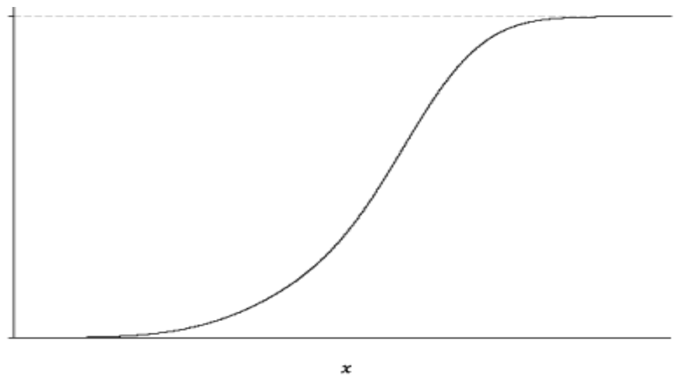
1. $F(x)$ is defined for all real numbers x.

2. $F(x)$ is non-decreasing. This follows easily: For any two real numbers a, b with $a \leq b$,
$$0 \leq P(a < X < b) = F(b) - F(a)$$
implying $F(a) \leq F(b)$.

3. $\lim_{x \to -\infty} F(x) = 0$

4. $\lim_{x \to \infty} F(x) = 1$. Conditions 3 and 4 imply that F is bounded between 0 and 1, that is, for all x, we have $0 \leq F(x) \leq 1$.

Based on these properties, we conclude that the general appearance of the cumulative distribution function of a continuous random variable is



Formally, a continuous random variable is a random variable whose cumulative distribution function is continuous everywhere. For, if there were any gaps in the cdf, say at a point x_0, then

$$P(X = x_0) = F(x_0 +) - F(x_0 -) \neq 0$$

Contradicting the fact $P(X = x_0) = 0$

So now, we have two methods of finding probabilities:

$$P(X \leq a) = P(X < a) = \int_{-\infty}^{a} f(x)dx = F(a)$$

$$P(X \geq a) = P(X > a) = \int_{a}^{\infty} f(x)dx = 1 - F(a)$$

$$P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b) = \int_{a}^{b} f(x)dx = F(b) - F(a)$$

EXAMPLES

Example 1. Let X be a continuous random variable and let $f(x) = 5x^4$ in $[0,1]$.

(a) Determine the cumulative distribution function $F(x)$

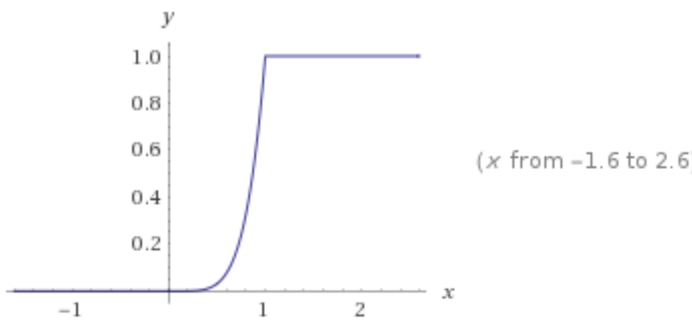
In the interval $[0, 1]$

$$F(x) = \int_{-\infty}^{x} f(t)\, dt = \int_{0}^{x} 5t^4 dt = [t^5]_0^x = x^5$$

Thus,

$$F(x) = \begin{cases} 0 & if \;\; x < 0 \\ x^5 & if \; 0 \le x \le 1 \\ 1 & if \; x > 1 \end{cases}$$

The graph of this cdf is given below:



(b) Find $P(X < 0.7)$

$$P(X < 0.7) = F(0.7) = (0.7)^5 = 0.168$$

(c) Find $P(X > 0.2)$

$$P(X > 0.2) = 1 - F(0.2) = 1 - (0.2)^5 = 0.999$$

(d) Find $P(0.3 \le X < 0.6)$

$$P(0.3 \le X < 0.6) = F(0.6) - F(0.3) = (0.6)^5 - (0.3)^5 = 0.075$$

Example 2. Let X be a continuous random variable and let $f(x) = \frac{2}{x^3}$ in $[1, \infty)$.

(a) Determine the cumulative distribution function $F(x)$

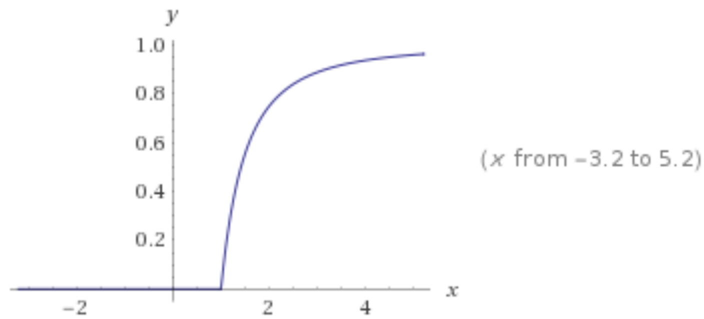
In the interval $[1, \infty]$

$$F(x) = \int_{-\infty}^{x} f(t)\, dt = \int_{1}^{x} \frac{2}{t^3} dt = \left[-\frac{1}{t^2}\right]_1^x = 1 - \frac{1}{x^2}$$

Thus,

$$F(x) = \begin{cases} 0 & if\ x < 1 \\ 1 - \dfrac{1}{x^2} & if\ x \geq 1 \end{cases}$$

The graph of this cdf is given below:



(b) Find $P(X < 2)$

$$P(X < 2) = F(2) = 1 - \frac{1}{4} = \frac{3}{4}$$

(c) Find $P(X > 1.5)$

$$P(X > 1.5) = 1 - F\left(\frac{3}{2}\right) = \frac{1}{(3/2)^2} = \frac{4}{9}$$

(d) Find $P(2 \leq X \leq 3)$

$$P(2 \leq X \leq 3) = F(3) - F(2) = \left(1 - \frac{1}{3^2}\right) - \left(1 - \frac{1}{2^2}\right) = \frac{5}{36}$$

Example 3. Let X be a continuous random variable and let $f(x) = 2e^{-2x}$ in $[0, \infty)$.

(a) Determine the cumulative distribution function $F(x)$

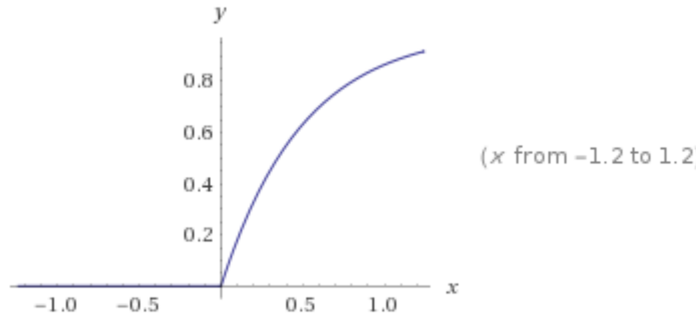
In the interval $[0, \infty]$

$$F(x) = \int_{-\infty}^{x} f(t)\, dt = \int_{0}^{x} 2e^{-2t}\, dt = [-e^{-2t}]_0^x = 1 - e^{-2x}$$

Thus,

$$F(x) = \begin{cases} 0 & if\ x < 0 \\ 1 - e^{-2x} & if\ x \geq\ 0 \end{cases}$$

The graph of this cdf is given below:



(b) Find $P(X < 1)$

$$P(X < 1) = F(1) = 1 - e^{-2} = 0.865$$

(c) Find $P(X > 1.5)$

$$P(X > 1.5) = 1 - F(1.5) = e^{-3} = 0.05$$

(d) Find $P(1 < X \leq 2)$

$$P(1 < X \leq 2) = F(2) - F(1) = (1 - e^{-4}) - (1 - e^{-2}) = e^{-2} - e^{-4} = 0.117$$

Example 4.

Suppose we have a cdf $F(x) = 1 - e^{-3x}$ in $[0, \infty)$ and is zero everywhere else. What is the pdf?

$$f(x) = \frac{dF}{dx} = \frac{d(1 - e^{-3x})}{dx} = 3e^{-3x}$$

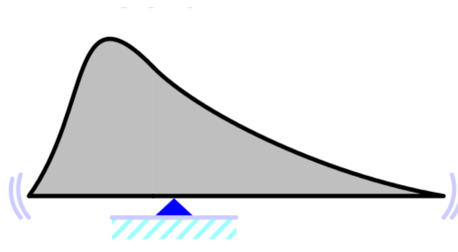
in $[0, \infty)$.

3 THE EXPECTED VALUE OF A CONTINUOUS RANDOM VARIABLE

The expected value of a continuous random variable can be computed in a similar fashion to that of a discrete random variable. All we have to do is to replace summation by integration. Thus, for a continuous random variable X with probability density function $f(x)$, the expected value is

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx$$

We still interpret the expected value as the center of mass.



EXAMPLES

Example 1. Let X be a continuous random variable and let $f(x) = 5x^4$ in $[0,1]$. Determine its expected value.

Since

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx$$

we have

$$E(X) = \mu = \int_0^1 x(5x^4)dx = \int_0^1 (5x^5)dx = \left[\frac{5}{6}x^6\right]_0^1 = \frac{5}{6}$$

Example 2. Let X be a continuous random variable and let $f(x) = \frac{2}{x^3}$ in $[1, \infty)$. Determine its expected value.

Since

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx$$

we have

$$E(X) = \mu = \int_{1}^{\infty} x\left(\frac{2}{x^3}\right) dx = \int_{1}^{\infty} \frac{2}{x^2} dx = \left[-\frac{2}{x}\right]_1^{\infty} = 2$$

Example 3. Let X be a continuous random variable and let $f(x) = 2e^{-2x}$ in $[0, \infty)$. Determine its expected value.

Since

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx$$

we have

$$E(X) = \mu = \int_{0}^{\infty} 2xe^{-2x}dx$$

Here, we apply integration by parts. Let $u = x$ and $dv = e^{-2x}dx$. Then

$$du = dx$$

and

$$v = -\frac{1}{2}e^{-2x}$$

Thus,

$$E(X) = \mu = 2\int_{0}^{\infty} xe^{-2x}dx = 2[xe^{-2x}]_0^{\infty} + 2 \times \frac{1}{2}\int_{0}^{\infty} e^{-2x}dx$$

Note that

$$2[xe^{-2x}]_0^{\infty} = 0 - 0 = 0$$

So,

$$E(X) = \mu = \int_{0}^{\infty} e^{-2x}dx = \frac{1}{2}$$

4 VARIANCE OF A CONTINUOUS RANDOM VARIABLE

As in the discrete case, the ***variance*** measures the dispersion or the variability in the distribution. Again, the formula is very similar to that of the variance of a discrete random variable. All we have to do is to replace summation by integration.

Let X be a continuous random variable with probability density function $f(x)$ and expected value $E(X) = \mu$. Then, the variance of X is given as

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$$

The standard deviation of X is

$$\sigma = \sqrt{V(X)}$$

As in the discrete case, we can rewrite the above formula in a form better suited for computational purposes.

Starting with the defining formula, let us actually take the square:

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx = \int_{-\infty}^{\infty} (x^2 - 2x\mu + \mu^2) f(x)dx$$

$$= \int_{-\infty}^{\infty} x^2 f(x)dx - \int_{-\infty}^{\infty} 2x\mu f(x)dx + \int_{-\infty}^{\infty} \mu^2 f(x)dx$$

Factoring out the constants, we get

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - 2\mu \int_{-\infty}^{\infty} x f(x)dx + \mu^2 \int_{-\infty}^{\infty} f(x)dx$$

Now, by definition

$$\int_{-\infty}^{\infty} x f(x)dx = E(X) = \mu$$

and

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Hence,

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - 2\mu^2 + \mu^2$$

Thus, we obtain the computational form of the formula for the variance as

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

This is the form we use in the following examples.

EXAMPLES

Example 1. Let X be a continuous random variable and let $f(x) = 5x^4$ in $[0,1]$. Recall, in this case, $E(X) = \mu = {}^5/_6$. Determine its variance.

Let us first compute

$$\int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^1 x^2(5x^4)dx = 5\int_0^1 x^6 dx = 5\left[\frac{x^7}{7}\right]_0^1 = \frac{5}{7}$$

Thus,

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2 = \frac{5}{7} - \frac{25}{36} = \frac{5}{252}$$

Example 2. Let X be a continuous random variable and let $f(x) = \frac{2}{x^3}$ in $[1, \infty)$. Recall, in this case, $E(X) = \mu = 2$. Determine its variance.

Let us first compute

$$\int_{-\infty}^{\infty} x^2 f(x)dx = \int_1^{\infty} x^2\left(\frac{2}{x^3}\right)dx = 2\int_1^{\infty} \frac{dx}{x} = 2[\ln(x)]_1^{\infty}$$

Since this expression → ∞, for this distribution the variance does not exist.

Example 3. Let X be a continuous random variable and let $f(x) = 2e^{-2x}$ in $[0, \infty)$. Recall, in this case, $E(X) = \mu = {}^1/_2$. Determine its variance.

Let us first compute

$$\int_{-\infty}^{\infty} x^2 f(x)dx = 2\int_0^{\infty} x^2 e^{-2x} dx$$

To this end, we will apply integration by parts twice. Let us denote this integral by I.

Put

$$u = x^2$$

and

$$dv = -\frac{1}{2}e^{-2x}dx$$

Thus,

$$I = 2\int_0^\infty x^2 e^{-2x}dx = 2\left(\left[-\frac{1}{2}x^2 e^{-2x}\right]_0^\infty + \frac{1}{2}\int_0^\infty 2xe^{-2x}dx\right)$$

Since

$$\left[-\frac{1}{2}x^2 e^{-2x}\right]_0^\infty = 0 - 0 = 0$$

$$I = \int_0^\infty 2xe^{-2x}dx$$

Apply integration by parts again. Let $u = x$ and $dv = e^{-2x}dx$. Then

$$du = dx$$

and

$$v = -\frac{1}{2}e^{-2x}$$

Thus,

$$I = \int_0^\infty 2xe^{-2x}dx = 2[xe^{-2x}]_0^\infty + 2 \times \frac{1}{2}\int_0^\infty e^{-2x}dx$$

Note that

$$[xe^{-2x}]_0^\infty = 0 - 0 = 0$$

So,

$$I = \int_0^\infty e^{-2x}dx = \frac{1}{2}$$

Thus,

$$V(X) = \sigma^2 = \int_{-\infty}^\infty x^2 f(x)dx - \mu^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

5 SPECIAL CONTINUOUS DISTRIBUTIONS

Now, we are going to study some special continuous distributions. Note that we are following the exact same order we did with discrete distributions.

The continuous distributions we will cover are

(i) Continuous Uniform Distribution
(ii) Exponential Distribution
(iii) Gamma Distribution
(iv) Weibull Distribution
(v) Beta Distribution
(vi) Normal Distribution

6 CONTINUOUS UNIFORM DISTRIBUTION

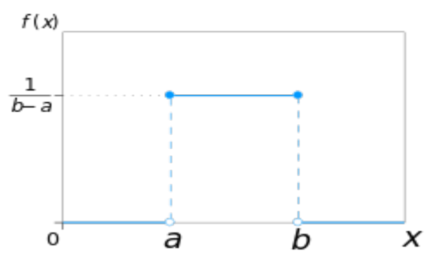
A ***continuous uniform distribution***, sometimes also known as a ***rectangular distribution***, is a distribution that has constant probability over an interval and is thus concerned with events that are equally likely to occur. Here is the formal definition

A continuous random variable X with probability density function

$$f(x) = \frac{1}{b-a}$$

for $a \leq x \leq b$, is said to have a ***continuous uniform distribution***.

The graph of $f(x)$ is, of course, a horizontal line over this interval.



Clearly, the area under $f(x)$ is the area of a rectangle with height $\frac{1}{b-a}$ and base $(b-a)$, which is

$$\frac{1}{(b-a)}(b-a) = 1$$

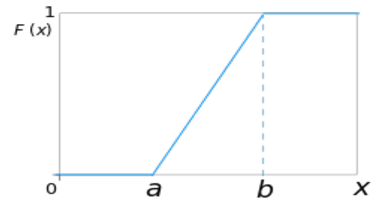
So, this is indeed a probability density function.

We can easily determine the cumulative distribution function $F(x)$. Over the interval $a \leq x \leq b$,

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_{a}^{x} \frac{1}{b-a}dt = \frac{1}{b-a}\int_{a}^{x} dt = \frac{x-a}{b-a}$$

Thus,

$$F(x) = \begin{cases} 0 & if\ x < a \\ \dfrac{x-a}{b-a} & if\ a \leq x \leq b \\ 1 & if\ x > b \end{cases}$$



We can easily determine the expected value.

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx = \int_{a}^{b} x\frac{1}{(b-a)}dx$$

$$= \frac{1}{(b-a)}\int_{a}^{b} xdx = \frac{1}{(b-a)}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{2}\frac{b^2-a^2}{b-a}$$

$$= \frac{b+a}{2}$$

Of course, this intuitively makes sense. The center of mass of a symmetric object will be at its center.

Similarly, we can find the variance. Recall that

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

Let us first find

$$\int_{-\infty}^{\infty} x^2 f(x)dx = \int_{a}^{b} x^2\frac{1}{b-a}dx =$$

$$\frac{1}{b-a}\int_{a}^{b} x^2 dx = \frac{1}{(b-a)}\left[\frac{x^3}{3}\right]_a^b = \frac{1}{3}\frac{b^3-a^3}{b-a}$$

$$= \frac{1}{3}\frac{(b-a)(b^2+ab+a^2)}{(b-a)} = \frac{1}{3}(b^2+ab+a^2)$$

Thus,

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

$$= \frac{1}{3}(b^2+ab+a^2) - \frac{1}{4}(a+b)^2 = \frac{(b-a)^2}{12}$$

Summary

For $a \leq x \leq b$, the pdf is

$$f(x) = \frac{1}{b-a}$$

The cdf, expectation and variance are:

$$F(x) = \begin{cases} 0 & if\ x < a \\ \dfrac{x-a}{b-a} & if\ a \leq x \leq b \\ 1 & if\ x > b \end{cases}$$

$$E(X) = \mu = \frac{b+a}{2}$$

$$V(X) = \sigma^2 = \frac{(b-a)^2}{12}$$

EXAMPLES

Example 1 The thickness of a flange on an aircraft component is uniformly distributed between 1 and 3 millimeters. What is the probability that thickness will be

Less than 1.7 millimeters?

Here $a = 1$ and $b = 3$. Thus, $f(x) = \frac{1}{2}$ over $1 \leq x \leq 3$. Hence, the required probability can be computed as the integral of $f(x)$, (which is, of course, the area of a rectangle with height $\frac{1}{2}$ and base 0.7):

$$\frac{1.7-1}{3-1} = 0.35$$

Alternately,

$$P(X < 1.7) = F(1.7) = \frac{1.7-1}{2} = 0.35$$

(a) Greater than 1.9 millimeters?

The required probability can be computed as the integral of $f(x)$, (which is, of course, the area of a rectangle with height $\frac{1}{2}$ and base 1.1)

$$\frac{3-1.9}{3-1} = 0.55$$

Alternately,

$$P(X > 1.9) = 1 - F(1.9) = 1 - \frac{1.9-1}{2} = 0.55$$

Example 2 Volume of a liquid in a container is uniformly distributed between 385 and 397 milliliters. What is the cumulative distribution function of this distribution? Use this function to find the probability that the volume is less than 392 milliliters.

Here $a = 385$ and $b = 397$. Thus,

$$F(x) = \begin{cases} 0 & if\ x < 385 \\ \dfrac{x-385}{12} & if\ 385 \le x \le 397 \\ 1 & if\ x > 397 \end{cases}$$

We have

$$P(X < 392) = F(392) = \frac{392-385}{12} = \frac{7}{12}$$

Example 3. The arrival time of a bus to a bus stop is uniformly distributed between 10 am and 10:30 am. If a person arrives at the bus stop at 10 am, what is the expected wait time?

The wait time is uniformly distributed between 0 minutes 30 minutes. Thus, $a = 0$ and $b = 30$. So,

$$E(X) = \mu = \frac{0+30}{2} = 15$$

Example 4. An adult can gain or lose two pounds in the course of a day. If weight change is uniformly distributed, what is the standard deviation of a person's weight over a day?

In this case, we have a uniform distribution with $a = -2$ and $b = 2$. Thus,

$$V(X) = \sigma^2 = \frac{(b-a)^2}{12} = \frac{16}{12} = \frac{4}{3}$$

Consequently,

$$\sigma = \sqrt{V(X)} = \sqrt{\frac{4}{3}} = 1.15$$

On a typical day, the change is about **1.155 lb away from the mean (0).**

7 THE EXPONENTIAL DISTRIBUTION

Let the random variable X denote the time (length) between the starting point and the first arrival in a Poisson process, i.e., a process in which events occur continuously and independently at a constant average rate λ (which is often called the ***rate parameter***). In this case, we say X has an ***exponential distribution***.

Since the exponential distribution is used to describe the lengths of the inter-arrival times in a Poisson process, it has many applications, such as

- the time until a radioactive particle decays
- the time it takes before we receive our next telephone call
- the time until default on a credit card payment
- the time between ER arrivals during a stable period (short windows)

and so on.

Given a Poisson process with rate λ, let the random variable N denote the number of arrivals in x units of time. Then, N has a Poisson distribution with mean λx. Assuming time (length) is not greater than x,

$$P(X > x) = P(N = 0) = \frac{e^{-\lambda x}(\lambda x)^0}{0!} = e^{-\lambda x}$$

Therefore,

$$F(x) = P(X \leq x) = 1 - P(X > x) = 1 - e^{-\lambda x}$$

Consequently, the probability density function would be

$$f(x) = F'(x) = \lambda e^{-\lambda x}$$

for $x \geq 0$.

Here is the formal definition:

The random variable X that equals the time (length) between successive events in a Poisson process with rate λ per unit interval is said to have an ***exponential distribution***. Its probability density function is

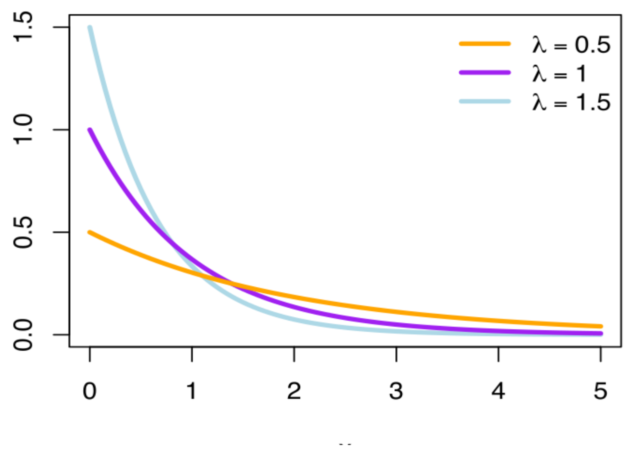
$$f(x) = \lambda e^{-\lambda x}$$

Here, λ is any positive real number, and $0 \leq x < \infty$.

This is indeed a probability density function since

$$\int_0^\infty \lambda e^{-\lambda x}\, dx = \left[-e^{-\lambda x}\right]_0^\infty = 0 - (-1) = 1$$

Below are graphs of exponential probability density function for various values of λ:



Using integration by parts, we can easily show that

$$E(X) = \mu = \frac{1}{\lambda}$$

and

$$V(X) = \sigma^2 = \frac{1}{\lambda^2}$$

Summary

For λ any positive real number, and $0 \leq x < \infty$, the pdf is

$$f(x) = \lambda e^{-\lambda x}$$

The cdf, expectation and variance are:

$$F(x) = 1 - e^{-\lambda x}$$

$$E(X) = \mu = \frac{1}{\lambda}$$

$$V(X) = \sigma^2 = \frac{1}{\lambda^2}$$

EXAMPLES

Example 1. During a decay, which is known to be a Poisson process, electrons hit a target at the rate 90 per hour. What is the probability that we will have to wait more than three minutes for the next electron to hit this target?

Here, $\lambda = {}^{90}/_{60} = 1.5$ per minute.

Thus, the probability of waiting more than three minutes will be

$$P(X > 3) = \int_3^\infty 1.5e^{-1.5x}dx = [-e^{-1.5x}]_3^\infty = 0 - (-e^{-4.5}) = e^{-4.5} = 0.011$$

Of course, since we know the cumulative distribution function, we could solve the same problem as

$$P(X > 3) = 1 - F(3) = 1 - (1 - e^{-4.5}) = e^{-4.5} = 0.011$$

Example 2. Log-ons to a computer system can be modeled as a Poisson process with 10 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes?

From the time there was a log-on, we want to have at least 6 minutes to pass before the next the log-on, in other words,

$$P(X > 6 \text{ min}) = P(X > 0.1 \text{ hr})$$

$$P(X > 0.1) = 1 - F(0.1) = 1 - \left(1 - e^{-10(0.1)}\right) = e^{-1} = 0.37$$

Example 3. The time between arrivals of buses at a bus stop is exponentially distributed with an expected value of 10 minutes. What is the probability we will wait less than 30 min?

Note that here we are given the expected value. Thus, $\lambda = \frac{1}{10} = 0.1$

$$P(X < 30) = \int_0^{30} 0.1e^{-0.1x}dx = [-e^{-0.1x}]_0^{30} = -e^{-3} - (-1) = 1 - e^{-3} = 0.95$$

Of course, since we know the cumulative distribution function, we could solve the same problem as

$$P(X < 30) = F(30) = (1 - e^{-3}) = 0.95$$

Example 4. The time between arrivals of buses at a bus stop is exponentially distributed with an expected value of 10 minutes. Determine the time t such that the probability you wait for less than t minutes is 0.3.

Note that here we are given the expected value. Thus, $\lambda = \frac{1}{10} = 0.1$
We want

$$P(X < t) = \int_0^t 0.1e^{-0.1x} dx = 0.3$$

This implies

$$[-e^{-0.1x}]_0^t = 1 - e^{-0.1t} = 0.3$$

Thus, we must have

$$e^{-0.1t} = 0.7$$

Taking natural logarithms on both sides

$$-0.1t = \ln(0.7) = -0.35667$$

implying

$$t \cong 3.6 \text{ min}$$

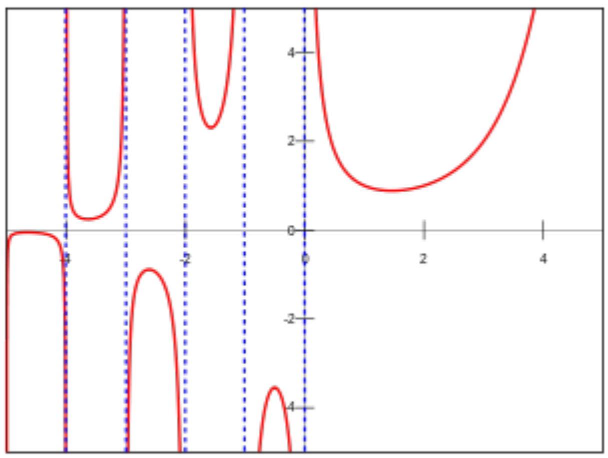
8 THE GAMMA FUNCTION

Before we discuss the ***gamma distribution***, let us take a brief detour and discuss the ***gamma function*** first.

Let t be a positive number. Then, the function defined by the formula

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$$

for $t > 0$, is called the ***gamma function***. The letter Γ is the capital gamma of the Greek alphabet. Below is the graph of the gamma function:



For small integer values of t we can compute the values of $\Gamma(\mathrm{t})$ by integration by parts. For example,

$$\Gamma(1) = \int_0^\infty e^{-x} dx = 1$$

$$\Gamma(2) = \int_0^\infty x e^{-x} dx = 1$$

$$\Gamma(3) = \int_0^\infty x^2 e^{-x} dx = 2$$

and so on. However, if we use this approach, to compute say $\Gamma(17)$, we need to apply the integration by parts formula seventeen times. Fortunately, there is a better way of doing this. To this end, we need a recursion formula.

A Recursion Formula

In

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$$

let $u = x^{t-1}$ and $dv = e^{-x} dx$ and apply integration by parts. Since we have

$$du = (t-1)x^{t-2}$$

and

$$v = -e^{-x}$$

we get

$$\Gamma(t) = [-x^{t-1}e^{-x}]_0^\infty + (t-1)\int_0^\infty x^{t-2} e^{-x} dx$$

Since

$$[-x^{t-1}e^{-x}]_0^\infty = 0 - 0 = 0$$

we have

$$\Gamma(t) = (t-1)\int_0^\infty x^{t-2} e^{-x} dx = (t-1)\,\Gamma(t-1)$$

Hence, we have the following recursion formula

$$\Gamma(1) = 1$$
$$\Gamma(t) = (t-1)\Gamma(t-1)$$

Thus, we can now easily compute

$$\Gamma(2) = 1 \times \Gamma(1) = 1$$
$$\Gamma(3) = 2 \times \Gamma(2) = 2$$
$$\Gamma(4) = 3 \times \Gamma(3) = 6$$
$$\Gamma(5) = 4 \times \Gamma(4) = 24$$

One observes the following very important rule:

If k is a positive integer, then

$$\Gamma(k) = (k-1)!$$

It can also be shown that

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Thus, using the recursion formula, we can also compute the values of the gamma function at halves of integers. For example,

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \times \Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi} \cong 0.89$$

As another example, let us compute $\Gamma\left(\frac{9}{2}\right)$.

$$\Gamma\left(\frac{9}{2}\right) = \frac{7}{2} \times \Gamma\left(\frac{7}{2}\right) = \frac{7}{2} \times \frac{5}{2} \times \Gamma\left(\frac{5}{2}\right)$$

$$= \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \Gamma\left(\frac{3}{2}\right)$$

$$= \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \Gamma\left(\frac{1}{2}\right) = \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2}\sqrt{\pi} \cong 11.63$$

We will use the gamma function when we discuss the gamma distribution (and also later when we discuss the Weibull distribution and the beta distribution).

9 THE GAMMA DISTRIBUTION

The relationship between the gamma distribution and the exponential distribution is analogous to the relationship between the geometric distribution and the negative binomial distribution.

We can think of the exponential distribution as the continuous analog of the geometric distribution and the gamma distribution as the continuous analog of the negative binomial distribution in the following sense:

- An exponential random variable describes the length until the **first event** is obtained in a Poisson process.
- A gamma random variable describes the length until ***r* events** occur in a Poisson process.

If X is the time until the rth event in a Poisson process, then we can show, (this would be a generalization of what we did in the case of the exponential distribution)

$$P(X > x) = \sum_{j=0}^{r-1} \frac{e^{-\lambda x}(\lambda x)^j}{j!}$$

This implies,

$$F(x) = 1 - P(X > x) = 1 - \sum_{j=0}^{r-1} \frac{e^{-\lambda x}(\lambda x)^j}{j!}$$

Differentiating and going through some algebraic simplifications, we obtain the following formal definition:

A random variable X that describes the time until the rth event in a Poisson process with average rate λ is called a ***gamma random variable***. Its probability density function is given as

$$f(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}$$

Here $0 \leq x < \infty, \lambda > 0$, and $r > 0$.

The parameters λ and r are called the ***rate parameter*** and the ***shape parameter***, respectively.

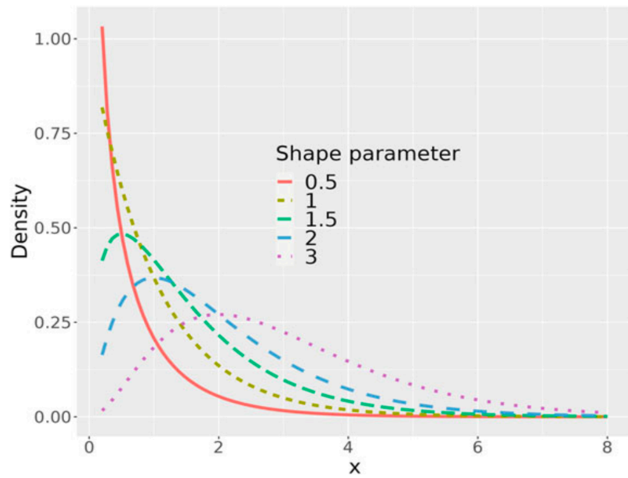
The expected value is given as

$$E(X) = \mu = \frac{r}{\lambda}$$

and the variance as

$$V(X) = \sigma^2 = \frac{r}{\lambda^2}$$

Below is an illustration of gamma densities for different values of the shape parameter r when $\lambda = 1$.



Since $\Gamma(1) = 1$, in case $r = 1$, these formulas reduce to

$$f(x) = \lambda e^{-\lambda x}$$

$$E(X) = \mu = \frac{1}{\lambda}$$

and the variance as

$$V(X) = \sigma^2 = \frac{1}{\lambda^2}$$

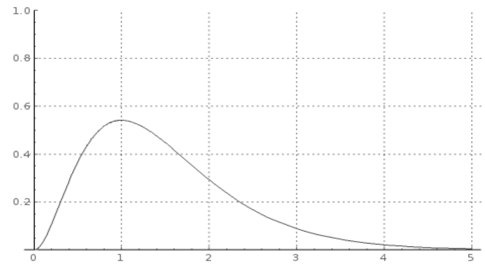
<u>In other words, for $r = 1$, the gamma distribution reduces to the exponential distribution.</u>

Also, note that, if r is a positive integer, then $\Gamma(r) = (r-1)!$, and the probability density function becomes

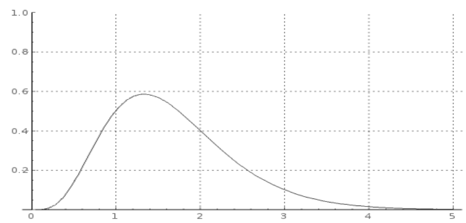
$$f(x) = \frac{\lambda^r}{(\mathrm{r}-1)!} x^{r-1} e^{-\lambda x}$$

Here $0 \leq x < \infty, \lambda > 0$, and r a positive integer.

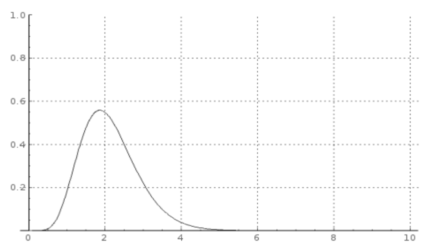
When we have this form, the distribution is referred to as the ***Erlang distribution***, named after Agner Krarup Erlang (1878 –1929), a Danish mathematician, statistician and engineer, who invented the fields of traffic engineering and queueing theory. Below are graphs of Erlang pdf for various values of r and λ



$r = 3$ and $\lambda = 2$



$r = 5$ and $\lambda = 3$



$r = 8$ and $\lambda = 3.75$

So, gamma distribution is a generalization of both the exponential and the Erlang distribution. Also, the ***chi-squared distribution*** (we will not see this distribution in this course) is a special case of the gamma distribution when $\lambda = {}^{1}\!/_{2}$ and r is one of the values ${}^{1}\!/_{2}$, 1, ${}^{3}\!/_{2}$, 2, …

The gamma distribution has a wide range of applications. A few are listed below.

- In wireless communication, it is used to model the multi-path fading of signal power
- In oncology, it models the age distribution of cancer incidence. In this case, the shape and scale parameters predict, respectively, the number of driver events and the time interval between them.
- In neuroscience, it is used to describe the distribution of inter-spike intervals

Summary of the Gamma Distribution

The pdf is

$$f(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}$$

Here $0 \leq x < \infty, \lambda > 0$, and , and $r > 0$.

The cdf has no expression in elementary functions when r is not a positive integer. The cdf when r is a positive integer, and the general expectation and variance and are:

$$F(x) = 1 - \sum_{j=0}^{r-1} \frac{e^{-\lambda x} (\lambda x)^j}{j!}$$

$$E(X) = \mu = \frac{r}{\lambda}$$

$$V(X) = \sigma^2 = \frac{r}{\lambda^2}$$

EXAMPLES

Example 1. Patients arrive at a hospital emergency center according to a Poisson process with mean of four per hour. What is the probability that more than 20 minutes is required for the second arrival?

In this case, $\lambda = 4,\ r = 2$ and $t =$ 20min $=$ 1/3hr. We want to compute $P(X > \frac{1}{3})$.

$$P\left(X > \frac{1}{3}\right) = \int_{1/3}^{\infty} \frac{4^2 x^1 e^{-4x}}{(1)!}\, dx = 16 \int_{1/3}^{\infty} x e^{-4x} dx$$

Let us apply integration by parts to $\int x e^{-4x} dx$ by setting $u = x$ and $dv = e^{-4x} dx$. Then,

$$du = dx$$

and

$$v = -\frac{1}{4} e^{-4x}$$

We get

$$\int x e^{-4x} dx = -\frac{1}{4} x e^{-4x} + \frac{1}{4} \int e^{-4x} dx = -\frac{1}{4} x e^{-4x} - \frac{1}{16} e^{-4x}$$

Multiplying by 16 and evaluating between $\frac{1}{3}$ and ∞, we get

$$0 + \left[\frac{4}{3} e^{-4/3} + e^{-4/3}\right] = 0.615$$

Example 2. The time delay in a system is approximated by a gamma distribution with expected value of one day and variance of 6 hours. What are the rate and shape parameters of this distribution?

Let us use “day” as our unit. We have

$$E(X) = \mu = \frac{r}{\lambda} = 1$$

and the variance as

$$V(X) = \sigma^2 = \frac{r}{\lambda^2} = \frac{1}{4}$$

From the second equation,

$$r = \frac{1}{4}\lambda^2$$

Thus, from the first equation

$$1 = \frac{r}{\lambda} = \frac{1}{4}\lambda$$

Consequently,

$$\lambda = 4$$

implying

$$r = 4$$

Hence both the scale parameter and the shape parameter is equal to 4.

10 THE WEIBULL DISTRIBUTION

Suppose we are interested in modeling the ***time to failure of a physical system***. In this case, we use a distribution called the ***Weibull distribution*** named after the Swedish mathematician/engineer Waloddi Weibull (1887-1979), who described it in detail in 1951. We should mention here that this distribution was in fact first identified by Fréchet in 1927.

Let the random variable X be the time to failure of a physical system. In this case, the probability density function is given as

$$f(x) = \frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1} e^{-\left(\frac{x}{\delta}\right)^{\beta}}$$

and we say X has a ***Weibull distribution***. Here, $0 \leq x < \infty$. The parameters δ and β can be any positive real numbers. We call δ the ***scale parameter*** and β the ***shape parameter***.

The scale parameter determines when, in time, a given portion of the population will fail. The shape parameter can be interpreted as follows:

A value of $\beta < 1$ indicates that the failure rate decreases over time

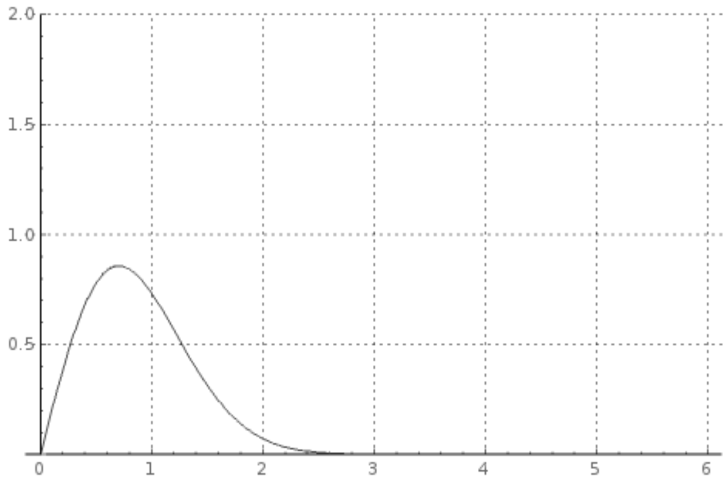
A value of $\beta = 1$ indicates that the failure rate is constant over time

A value of $\beta > 1$ indicates that the failure rate increases with time

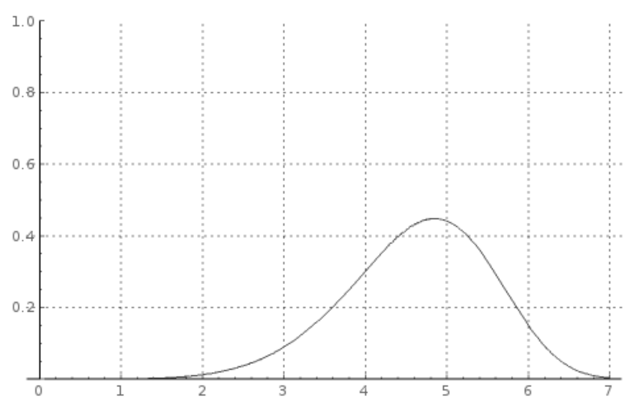
The Weibull distribution has a wide range of applications, some of which are given below:

- In electrical engineering to represent overvoltage occurring in an electrical system
- In industrial engineering to represent manufacturing and delivery times
- In weather forecasting to describe wind speed distributions
- In communications systems engineering
- In radar systems to model the dispersion of the received signals level produced by some types of clutters

Below are the graphs of some Weibull distributions:



$\beta = 2, \delta = 1$



$\beta = 6, \delta = 5$

Special Cases

In case $\beta = 1$, the Weibull distribution reduces to an exponential distribution with $\lambda = 1/\delta$

In case $\beta = 2$, the Weibull distribution is called the Rayleigh distribution.

We can find the cumulative distribution function in a relatively straightforward way. In this case,

$$F(x) = \int_{-\infty}^{x} f(t)dt = \int_{0}^{x} \frac{\beta}{\delta}\left(\frac{t}{\delta}\right)^{\beta-1} e^{-\left(\frac{t}{\delta}\right)^{\beta}} dt$$

Put

$$u = \left(\frac{t}{\delta}\right)^{\beta}$$

Then

$$du = \frac{\beta}{\delta}\left(\frac{t}{\delta}\right)^{\beta-1} dt$$

and the above integral becomes

$$\int e^{-u} du = -e^{-u}$$

implying

$$F(x) = 1 - e^{-\left(\frac{x}{\delta}\right)^{\beta}}$$

It can also be shown that

$$E(X) = \mu = \delta\Gamma\left(1 + \frac{1}{\beta}\right)$$

and

$$V(X) = \sigma^2 = \delta^2\,\Gamma\left(1 + \frac{2}{\beta}\right) - \mu^2$$

Summary

The pdf is

$$f(x) = \frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1} e^{-\left(\frac{x}{\delta}\right)^{\beta}}$$

for $0 \leq x < \infty$.

The cdf, expectation and variance are:

$$F(x) = 1 - e^{-\left(\frac{x}{\delta}\right)^{\beta}}$$

$$E(X) = \mu = \delta\Gamma\left(1 + \frac{1}{\beta}\right)$$

$$V(X) = \sigma^2 = \delta^2\,\Gamma\left(1 + \frac{2}{\beta}\right) - \mu^2$$

EXAMPLES

Example 1. The time to failure of a bearing in a mechanical shaft is modeled by a Weibull distribution with $\beta = \frac{1}{3}$ and $\delta = 4000$ hours. What is the probability the system will last for more than 5700 hours?

$$P(X > 5700) = 1 - F(5700) = 1 - \left(1 - e^{-\left(\frac{5700}{4000}\right)^{\frac{1}{3}}}\right) = e^{-1.125} = 0.325$$

Example 2. Ocean wave heights are described using a Weibull distribution with expected value 2.5 m and shape parameter 2. What is the standard deviation of this distribution?

$$\delta\Gamma\left(1 + \frac{1}{\beta}\right) = 2.5$$

$$\beta = 2$$

This implies,

$$\delta\Gamma\left(\frac{3}{2}\right) = 2.5$$

$$\frac{1}{2}\delta\sqrt{\pi} = 2.5$$

$$\delta = \frac{5}{\sqrt{\pi}} = 2.82$$

Thus, the variance is

$$\delta^2\,\Gamma\left(1 + \frac{2}{\beta}\right) - \mu^2 = (2.82)^2\Gamma(2) - (2.5)^2 = 1.71$$

Consequently,

$$\sigma = \sqrt{V(X)} = \sqrt{1.71} = 1.31$$

Example 3. Lifetime of a component (in hours) is modeled by a Weibull distribution with $\beta = 2$ and $\delta = 5000$ hours. Given that the system has lasted for more than 3000 hours, what is the probability it will last more than 8000 hours?

Let A be the event the system lasts for 8000 hours and let B be the event it last for more than 3000 hours. We want to compute $P(A|B)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Note that in this case $A \cap B = A$.

Hence the required probability is

$$\frac{P(A)}{P(B)} = \frac{1 - F(8000)}{1 - F(5000)} = \frac{e^{-2.56}}{e^{-1}} = e^{-1.56} = 0.21$$

11 THE BETA FUNCTION

The ***beta function*** (also called the ***Euler integral of the first kind***) is a function of two variables defined as

$$\mathrm{B}(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1}\ dt$$

So for example,

$$\mathrm{B}(2,1) = \int_0^1 t\ dt = \left[\frac{1}{2}t^2\right]_0^1 = \frac{1}{2}$$

The beta function is closely related to the gamma function. In fact, it can be shown that

$$\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

Thus, if x, y are positive integers

$$\mathrm{B}(x, y) = \frac{(x-1)!\,(y-1)!}{(x+y-1)!}$$

So,

$$\mathrm{B}\left(\frac{1}{2}, 2\right) = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma(2)}{\Gamma\left(\frac{5}{2}\right)} = \frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{5}{2}\right)} = \frac{\Gamma\left(\frac{1}{2}\right)}{\frac{3}{2}\times\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)} = \frac{4}{3}$$

$$\mathrm{B}(5,3) = \frac{4!\,2!}{7!} = \frac{1}{105}$$

$$\mathrm{B}(9,2) = \frac{8!\,1!}{10!} = \frac{1}{90}$$

$$\mathrm{B}(7,5) = \frac{6!\,4!}{11!} = \frac{1}{330}$$

12 THE BETA DISTRIBUTION

The ***beta distribution*** is a continuous distribution defined over a finite interval (almost always $[0,1]$) and is therefore useful to model probabilities or proportions. In other words, **<u>the beta distribution is a probability distribution *on probabilities*</u>**.

Some simple applications involve

- Proportion of solar radiation absorbed by a material
- Proportion of the maximum time allotted needed to finish a project
- The initial allele proportions in a genetic simulation

Let us give a formal definition.

A random variable X with probability density function

$$f(x) = \frac{1}{\mathrm{B}(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1} = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

for $0 \leq x \leq 1$, is called a ***beta distribution with parameters α and β***, where α and β are any positive real numbers and are called ***shape parameters***.

We do not have a closed form for the cumulative distribution function of a beta distribution, so to compute probabilities we will have to integrate the probability density function within appropriate limits.

Let us try to interpret this pdf a bit more closely. It is of the form

constant × variable to a power × (1 − variable) to a power

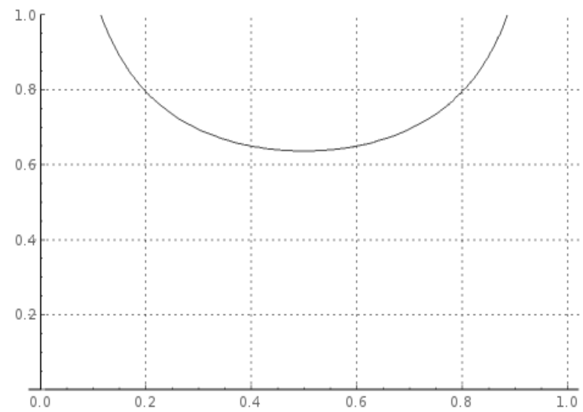
Actually, we have seen a function of this form before, namely the pmf of a ***binomial distribution***:

$$\binom{n}{x} p^x (1-p)^{n-x}$$

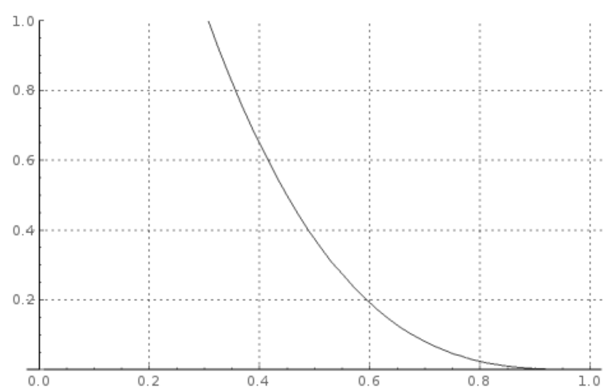
In case of binomial distribution, we are given the probability of success p and we are interested in computing probability of a certain number of successes, x. However, in case of beta distribution we are trying to compute the probability of having a certain probability of success.

This approach also provides us a means of interpreting the parameters α and β. We can think of $(\alpha - 1)$ as the number successes and $(\beta - 1)$ as the number of failures just like the powers x and $(n - x)$ in the binomial distribution. So, if we are free to set the parameters α, β in an application that requires the use of a beta distribution, we should proceed as follows: If we believe the probability of success to be high, we should choose α to be larger than β, and vice versa.

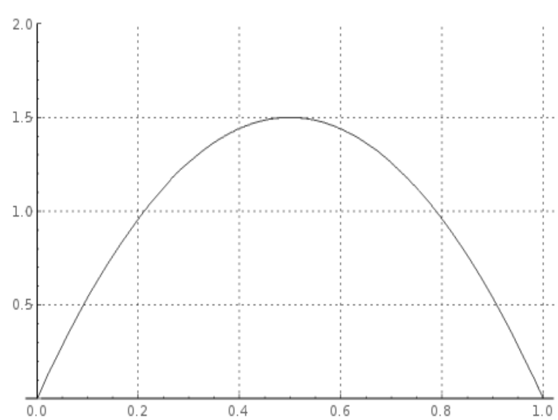
Below are graphs of the beta distribution for some values of α and β:



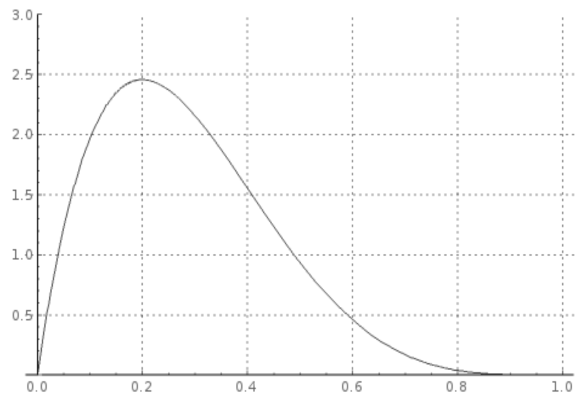
$\alpha = \beta = 0.5$



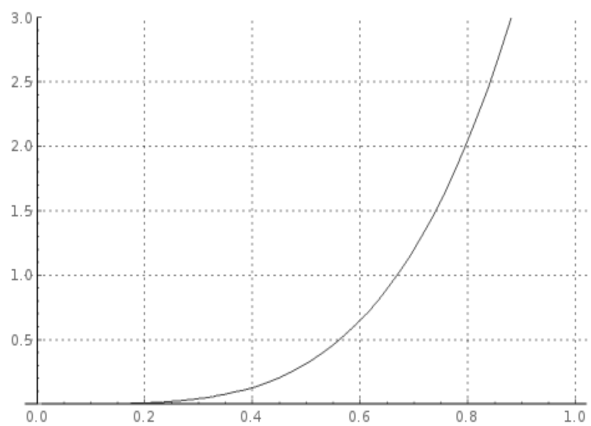
$\alpha = 1, \beta = 3$



$\alpha = \beta = 2$



$$\alpha = 2, \beta = 5$$



$$\alpha = 5, \beta = 1$$

By definition of the beta function and a rather straightforward application of the recursion formula for the gamma function, we can show that

$$E(X) = \mu = \frac{\alpha}{\alpha + \beta}$$

and

$$V(X) = \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

We can also show that the maximum value of the probability density function (called the ***mode***), is always in the interior of $[0, 1]$ if $\alpha, \beta > 1$.

Recall that to determine the maximum (or minimum) values of a function $f(x)$, one solves the equation $f'(x) = 0$.

Since

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

$$\begin{aligned} f'(x) &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\Big((\alpha-1)x^{\alpha-2}(1-x)^{\beta-1} - (\beta-1)x^{\alpha-1}(1-x)^{\beta-2}\Big) \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\left(x^{\alpha-2}(1-x)^{\beta-2}\right)\big((\alpha-1)(1-x) - (\beta-1)x\big) \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\left(x^{\alpha-2}(1-x)^{\beta-2}\right)\big((\alpha+\beta-2)x - (\alpha-1)\big) \end{aligned}$$

Now let us solve the equation $f'(x) = 0$.

The constant $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$ cannot be zero. Since we are looking for the maxima in the interior of $[0,1]$ $x \neq 0$ and $(1-x) \neq 0$. Thus the only term that can be equal to zero is

$$(\alpha+\beta-2)x - (\alpha-1) = 0$$

Hence, the mode (maximum) occurs at

$$x = \frac{\alpha-1}{\alpha+\beta-2}$$

Summary

The cdf, expectation and variance are:

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

$$E(X) = \mu = \frac{\alpha}{\alpha+\beta}$$

$$V(X) = \sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Mode is at

$$x = \frac{\alpha-1}{\alpha+\beta-2}$$

EXAMPLES

Example 1. Suppose the proportion of the total service time in a disassembly project follows a beta distribution with $\alpha = 3.5$ and $\beta = 2$. What is the probability that total service time will exceed 70% of the maximum time allowed?

The probability distribution function is

$$f(x) = \frac{\Gamma(5.5)}{\Gamma(3.5)\Gamma(2)} x^{2.5}(1-x)$$

By recursion formula,

$$f(x) = \frac{4.5 \times 3.5\Gamma(3.5)}{\Gamma(3.5)\Gamma(2)} x^{2.5}(1-x)$$

$$= \frac{63}{4}(x^{5/2} - x^{7/2})$$

Thus, the required probability is

$$P(X > 0.7) = \frac{63}{4} \int_{0.7}^{1} \left(x^{5/2} - x^{7/2}\right) dx = \frac{63}{4}\left[\frac{2}{7}x^{7/2} - \frac{2}{9}x^{9/2}\right]_{0.7}^{1}$$

$$= \left[\frac{9}{2}x^{7/2} - \frac{7}{2}x^{9/2}\right]_{0.7}^{1} = 1 - (0.58830) = 0.412$$

Example 2. The proportion of solar radiation that enters a room through a low emission window glazing is a beta distribution with $\alpha = 5$ and $\beta = 3$. What is the probability that less than 60% of the solar radiation will enter the room?

The probability distribution function is

$$f(x) = \frac{\Gamma(8)}{\Gamma(5)\Gamma(3)} x^4(1-x)^2$$

$$= \frac{7!}{4!\,2!} x^4(1-x)^2$$

$$= 105\, x^4(1 - 2x + x^2)$$

$$= 105\, (x^4 - 2x^5 + x^6)$$

We want to compute

$$P(X < 0.6) = 105 \int_0^{0.6} (x^4 - 2x^5 + x^6)\, dx = 105 \left[\frac{x^5}{5} - \frac{x^6}{3} + \frac{x^7}{7}\right]_0^{0.6}$$

$$= [21x^5 - 35x^6 + 15x^7]_0^{0.6} = 0.42$$

Example 3. Find the expected value, the variance, the standard deviation, and location of the mode for the distribution given in problem 1.

$$E(X) = \mu = \frac{\alpha}{\alpha + \beta} = \frac{3.5}{5.5} = 0.64$$

$$V(X) = \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{7}{(5.5)^2(6.5)} = 0.036$$

$$\sigma = \sqrt{V(X)} = 0.19$$

Mode is at

$$x = \frac{\alpha - 1}{\alpha + \beta - 2} = \frac{2.5}{3.5} = 0.714$$

Example 4. Find the expected value, the variance, the standard deviation, and location of the mode for the distribution given in problem 2.

$$E(X) = \mu = \frac{\alpha}{\alpha + \beta} = \frac{5}{8} = 0.625$$

$$V(X) = \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{15}{(8)^2(9)} = 0.026$$

$$\sigma = \sqrt{V(X)} = 0.16$$

Mode is at

$$x = \frac{\alpha - 1}{\alpha + \beta - 2} = \frac{4}{6} = 0.67$$